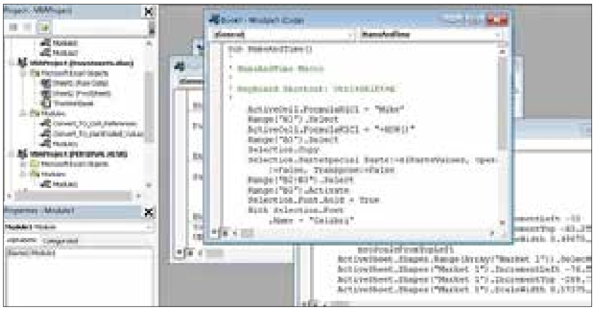
Twitter tests new features on Twttr
Twitter is testing threaded replies and more such features on its prototype app, Twttr. **https://dgit.in/dec19-43**

Flipkart, Nokia to launch 4K TVs
Flipkart and Nokia are teaming up to lauch 4K smart TVs using, for the first time, JBL audio technology. **https://dgit.in/dec19-44**



Excel VBA in Action

can make use of the integrated commands to create macros. Emacs is being developed based on LISP programming language, which offers a deep extension capability. You will be able to write new commands for the editor. You can develop macros for the management of email, files, RSS feeds and much more.

Vim

Vim was developed by Bram Moolenaar for the Amiga computer in 1988. The first release Vim v1.14 was released in 1991. Vim was originally named as Vi IMitation but was modified to Vi IMproved in 1993. Vim was developed by drawing inspiration from Stevie developed by Tim Thompson for the Atari ST in 1987.

Early LISP macros

Before the arrival of macros, LISP featured FEXPRs, which is the function-specific operator. The inputs are syntactic forms of the arguments instead of the computed arguments. The FEXPRs was implemented at the same level as EVAL. Timothy Hart announced the addition of macros in 1963 to LISP 1.5 in AI Memo 57: MACRO Definitions for LISP.

Anaphoric macros

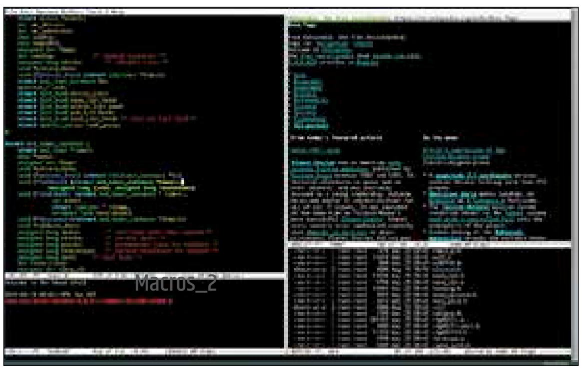
An Anaphoric macro deliberately captures the relevant form supplied to the macro. The term Anaphor means an expression that refers to another. The Anaphoric macro was introduced in Paul Graham's On Lisp.

Hygienic macro

Hygienic macros are macros that are assured not to cause the accidental capture of identifiers. These macros are a programmatic solution to the capture issue integrated with the macro expander. The term hygienic was drafted in the 1986 paper authored by Kohlbecker and Wand in the form of KFFD algorithm. The Syntactic closures evolved as an alternative by Bawden and Rees in 1988. Dybvig introduced the syntax-case macro system in 1993. The languages that provide support for hygienic macro mechanism are Scheme, Racket, Nemerle, Dylan, Elixir, Nim, Rust, Haxe, Mary2, Julia and Raku. Some of the competing implementations of hygienic macros are syntax-rules, syntax-case, explicit renaming and syntactic closures.

Parameterized macro

The purpose of the Parameterized macro is to insert the specified objects into its expansion < #define pred(x) ((x)-1) > You can pass any value to the argument 'x'. The parameterized macros are mainly used in C, LISP, PL/I and Scheme. The computer typesetting system TeX makes extensive use of macros. MacroML is used to reconcile static typing and macro systems. Some of the other macro scripting languages are troff, nroff, CMS EXEC, EXEC 2, CLIST, REXX and SCRIPT.



Emacs unleashes the full potential of the macros

Alternatives to Macros

Macros are aggressively used by Microsoft Excel users to perform complex mathematical operations. Even though you can create macros from within Excel, the complete dependency of Excel will be harmful in the long run. The best alternative is to make use of open-source platforms such as Python, PHP and R that can be used to perform any complex tasks. This includes aggregate, reshape, pivot, graph and plot including the ability to import and export data into databases. You can also make use of lists and arrays to create packages instead of buggy DLLs and COMs. Moreover, consoles such as PyCharm and RStudio stand out of the crowd with powerful IDE when compared with Visual Studio. If you are not interested to work with any of these solutions, you can try exclusive Robotic process automation software available on the market.

Tools for Macros Development

You can create macros using third-party software programs such as Macro Express (https://dgit.in/macroexp), Macro Recorder (https://dgit.in/macrorec), Macro Recorder (https://dgit.in/macrorecor), AutoHotkey (https://dgit.in/autohotkey), Quick Macros (https://dgit.in/quickmacros), Macro Toolworks (https://dgit.in/macrotool), AutoIt (https://dgit.in/autoit), Auto Mouse Clicker (https://dgit.in/automouseclicker), EasyClicks (https://dgit.in/easyclicks), Pulover's Macro Creator (https://dgit.in/puloversmacro) and ShortKeys (https://dgit.in/shortkeys).

CONCLUSION

With the help of Macros, programmers will be able to avoid repetitive coding completely. The effective use of Macros will enable developers not only to reap rich dividends but also to enhance productivity. With the rapid growth of technology, the development of macros have been simplified to a large extent with plenty of tools.